

## Exam

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This is an **individual exam**. You are required to submit, no later than May 7, by email (ytkpegli@univ-pau.fr), a zip folder containing:

- a PDF report presenting your answers to all questions;
- a do-file containing the STATA code used to obtain the results reported in the PDF.

### Exercise 1 - Out-of-Pocket Health Expenditures

The aim of this exercise is to model out-of-pocket health expenditures using the Guinea Living Standards Measurement Study 2018–19 (see the data supplied). To that end:

1. Read the contents of the dataset to identify which information is included, especially the out-of-pocket health expenditure (**oops\_sante**) and relevant explanatory variables.
2. Estimate OLS, Tobit, sample selection, and two-part models for out-of-pocket health expenditures.
3. Discuss the consistency and efficiency of each estimator.

### Exercise 2 - Speed of convergence

The aim of this exercise is to estimate the speed of convergence postulated in Solow-style models of economic growth. To that end, you need to:

1. Download data from an online platform of your choice on GDP growth and relevant control variables, as suggested by Solow-style models
2. Estimate a dynamic panel model using pooled OLS, the random effects estimator, the within estimator, the first-difference estimator, GMM in first differences, and system GMM.
3. Discuss the consistency and efficiency of each estimator.